

1) Human-AI Collaborative Workflows with LLMs/VLMs

Emerging research has studied the application of LLMs and VLMs to infrastructure inspection, with initial systems demonstrating natural language interaction with infrastructure digital twins. Similarly, existing work appears primarily aimed at developing new systems, including damage detection and description, report generation, and autonomous UAV operation, with human-centred factors remaining less studied. However, human expertise remains a crucial component in the inspection process, providing valuable oversight and contextual knowledge beyond what is captured in the system. Furthermore, while natural language interaction via LLMs provide an intuitive way of interacting with multimodal data, it carries a variety of risks, such as producing convincing but incorrect assessments, and users relying on these assessments without further verification.

Building reliable human-AI collaborative workflows that understands and optimises the interaction between the system, multimodal data, and LLMs is currently described as a long-term research challenge in the literature. Similarly, user evaluations in general and relevant concepts such as cognitive load and user experience appear to be generally little explored that would contribute towards making a system usable in the long-term.

The goal is to establish groundwork towards a collaborative human-AI system for bridge inspection that supports and augments human cognitive processes, enhance trustworthiness, and increase decision efficiency by enabling effective intervention via the following research question:

How should a collaborative workflow between experts and the system present information and structure interaction in order to support expert decision-making?

Considering the following dimensions:

- **Trust Calibration:** Whether the user's trust matches the system's actual reliability in a given instance. Appropriate trust assessment allows the human expert to apply their own knowledge when they identify that the model is not performing well for better decision outcomes.
- **Appropriate Reliance:** Acceptance of correct output and rejection of incorrect output. Calibrated trust supports appropriate reliance (Effectively decision behaviour).
- **Explainability and transparency:** Presenting how the data was produced, what data it draws on and the evidence it refers to. Intended to support trust calibration and appropriate reliance.

Some relevant papers:

Çelik, R. F., Bağcı, U., & Kessler, S. (2026). Vision language models for bridge inspections: A review of applications in image-based damage documentation. *Structural Concrete*.

(does not look at particular human-AI collaboration metrics but presents a user evaluation)

Luo, P., Wen, X., Jiang, Y., Pärn, E., & Brilakis, I. (2026). ChatTwin: Bridging the usability gap to digital twin adoption in infrastructure operations and maintenance with a Natural Language Interface. *Automation in Construction*, 181, 106655.

Raees, M., Khan, V. J., Lykourantzou, I., & Papangelis, K. (2026, April). Do People Appropriately Rely on AI-Advice? An Analytical Review of HCI Research on Human-AI Decision-Making. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (pp. 1-24).

2) Interaction Modality with XR

Extended reality seems to have been explored as an interface to digital twins in construction and infrastructure quite a bit. Not very deep yet in the literature regarding digital twins, but some potential avenues in addition to 1)., though that would need stronger literature support.

- Integrating digital twin in VR with the conversational agent, whether the spatiality of the presentation improves workflow and decision outcome / which aspects of the workflow could be improved.
- Planning drone flight path - e.g. gestures coupled with voice commands to plan/adjust a flight path, increase situational awareness
- Situated presentation of information in VR / adaptation of information presentation depending on the medium used
- Use of AR for on-site use (AR can be simulated) especially considering the new wave of AR glasses with potential for better adoption: presenting and interacting with data from the digital twin on-site. Alternatively smartglasses, challenge would be the limited field of view.

Some (somewhat) relevant papers:

Marzban, M., Sarvesh, M., Maroun, C., Wells, B., Wang, N., Nam, H., ... & Kim, K. (2025, October). An Immersive Digital Twin with Virtual Agent Interface for Pipeline Leak Simulation and Monitoring. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 201-204). IEEE.

Smith, A., Duff, C., Sarlo, R., & Gabbard, J. L. (2022, March). Wearable augmented reality interface design for bridge inspection. In *2022 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops (VRW)* (pp. 497-501). IEEE.